ECEN 104 – Lab 1: Resistor Tolerance and Probability of Success

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Introduction

For this lab, we are simulating being engineers for XYZ Corp tasked with designing a circuit for a NASA moon lander. This lab specifically investigates the probability that a batch of 1% tolerance resistors will enable a successful build of a design. Our design only required ten resistors, but all had to work, or the system would fail.

We, along with other members of our lab, collected resistance measurements using the digital multimeters in the lab and combined our data in order to perform statistical analysis (assuming that these findings will extrapolate to future parts of this type). We decided that each group should upload their resistance measurements to a shared Google Sheet, which was then downloaded as a .CSV file and imported into MATLAB via the readtable() function. The data was then converted into a vector, and further analysis was performed.

By measuring resistor values and comparing them to both manufacturer tolerance limits and system requirements, we calculated the probability that a single resistor will function correctly. We then extended this analysis to determine the probability that an entire system requiring ten resistors will succeed. Our findings reveal that while single resistors often meet system requirements, the combined probability of success for multiple resistors drops significantly.

Theory

For nominal 100 kΩ resistors with a 1% tolerance, the manufacturer specifies an acceptable range of 99 kΩ to 101 kΩ. However, system-level functionality may tolerate a different range. In this lab, the system requirement was set to 98.5–100.5 kΩ. This highlights that tolerance failures do not necessarily equate to system failures.

From the dataset of 216 resistors, we determined the probability of a resistor being acceptable. The probability of system failure was calculated as 35.65%, giving a success probability of 64.35% per resistor. Because the design requires ten resistors to all function within the acceptable range, the overall probability of success is given by the product of individual probabilities:

P(success for 10 resistors) = (0.6435)^10 ≈ 1.22%.

Experimental Setup

A total of 216 resistors were measured with a digital multimeter, and their values were recorded in kilo-ohms (kΩ) to the tenths place. Each measurement was compared to both the manufacturer's tolerance window (99–101 kΩ) and the system requirement window (98.5–100.5 kΩ). The counts of out-of-spec resistors were then used to calculate

probabilities of failure and success. Finally, the probability for a ten-resistor system was determined by raising the single-resistor success probability to the 10th power.

Results

The dataset of 216 resistors yielded the following results:

- **Total resistors measured:** 216
- **Count outside 99–101 kΩ (manufacturer tolerance):** 95
- **Count outside 98.5–100.5 kΩ (system requirement):** 77

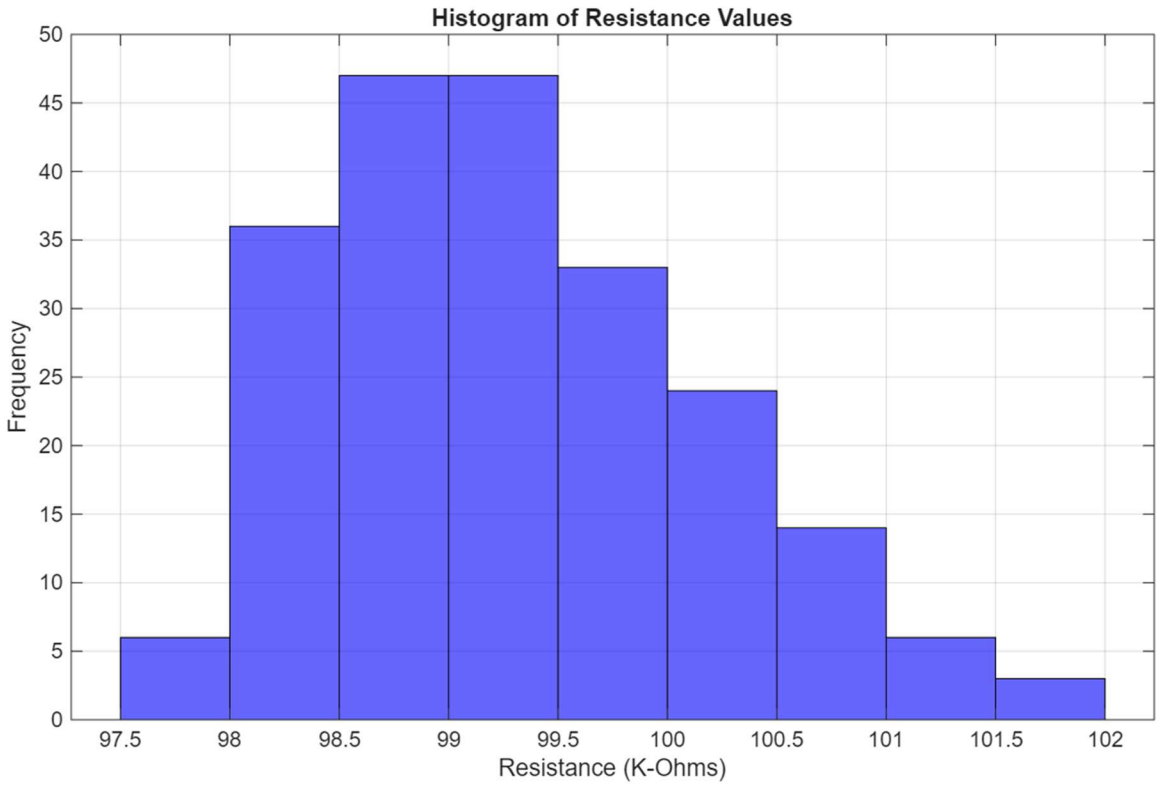
From this, we calculated:

- **Probability of tolerance failure:** 43.98%
- **Probability of system failure (single resistor):** 35.65%
- **Probability of system success (single resistor):** 64.35%
- **Probability of system success (ten resistors):** 1.22%

Table 1 summarizes the results.

Evaluation Criterion	Probability
Tolerance failure (99–101 kΩ)	43.98%
System failure (single resistor, 98.5–100.5 kΩ)	35.65%
System success (single resistor)	64.35%
System success (ten resistors)	1.22%

Figure 1 shows the distribution calculated in MATLAB.



Discussion

While nearly 44% of resistors fell outside manufacturer specifications, only 27% failed to meet system requirements. This indicates that tolerance alone is not a sufficient measure of functional success. However, when scaling from one resistor to a system requiring ten, the cumulative probability of success falls dramatically: 64.35% for a single resistor to just 4.39% for ten resistors. This illustrates the compounding effect of requiring multiple independent components to all meet specifications.

The distribution of resistor values is roughly bell-shaped but slightly skewed toward lower resistances, with most values clustering between 98.5 kΩ and 99.5 kΩ. Only a small number extend into the higher range above 101 kΩ, creating a longer tail on the low side. The slight left skew is beneficial for our design, as our design can tolerate a relatively lower resistance when compared to the average vs a higher one. However, this slight benefit still did not help to create a good chance of success.

Potential sources of error include measurement uncertainty from multimeter error, resistor batch variation, and environmental factors such as temperature and humidity.

Conclusion

This lab showed that while individual 1% tolerance resistors had a 64.35% chance of meeting the system requirements, the probability that all ten resistors would

simultaneously succeed was only 1.22%. This stark difference demonstrates how system reliability depends on the compounded probabilities of its individual components. We do not recommend NASA to proceed with the mission, and us engineers at XYZ Corp need to either make a new circuit with more failure tolerance or find a new resistor vendor.